

Effects of Sensor Size on the image

The sensor with the larger individual pixels will record the light it receives with greater precision than those with smaller pixels. Larger the sensor, larger are the pixels on it, implying that they are more receptive and sensitive to light and can therefore produce more detailed images.

An 8 megapixel Point and Shoot camera has the same number of pixels as an 8 megapixel DSLR camera. But the DSLR has a larger sensor with larger pixels than the teeny-weenie sensor on your point and shoot.

These pixels can collect more light, recording the highs and lows of the intensity of light at each pixel more accurately, reducing noise and bettering detail and image quality.

Being more sensitive to light, the larger image sensor can give you much better pictures in low light conditions than the standard P&S ones.

If the sensor is smaller than actual requirement, then the camera is unable to capture the full view of the image.

The smaller the sensor the more the pixels have to be crammed into the space available. Diffraction, that is the spreading of light when it passes through an aperture, spoils the image, leading to 'noise'.

Noise level is directly connected to the actual size of the pixels on a sensor and while several noise reduction techniques are being used in modern digital cameras, the fact remains that larger sensors are liable to produce less noisy photographs. The downside here is that as the pixel size increases, so does the size of the file – thus requiring more memory space to be stored in – and longer processing time.

A large sensor, however would give the same resolution as a small one if both of them have the same number of pixels. Pixel size does not affect resolution, which remains constant for the sensor count.

A 3-megapixel sensor, for example, would give you the same result, whatever the size of the sensor – whether it's the little digicam type or the big DSLR one.

It's the size of the pixels that makes all the difference. The table would help to elucidate what we've been talking about.